

Progress Report: Sprint 9

CS.028 Optimizing Research Software Codes - 18 February 2026

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Summary

- Completed Architecture and Risks Update.
- Adjusted parse() IO function for a 3% improvement.
- Vectorized to _mjd() to remove .apply() calls for a 2.5% improvement.

Sprint Work Log

Parse IO Adjustment

Status: Done

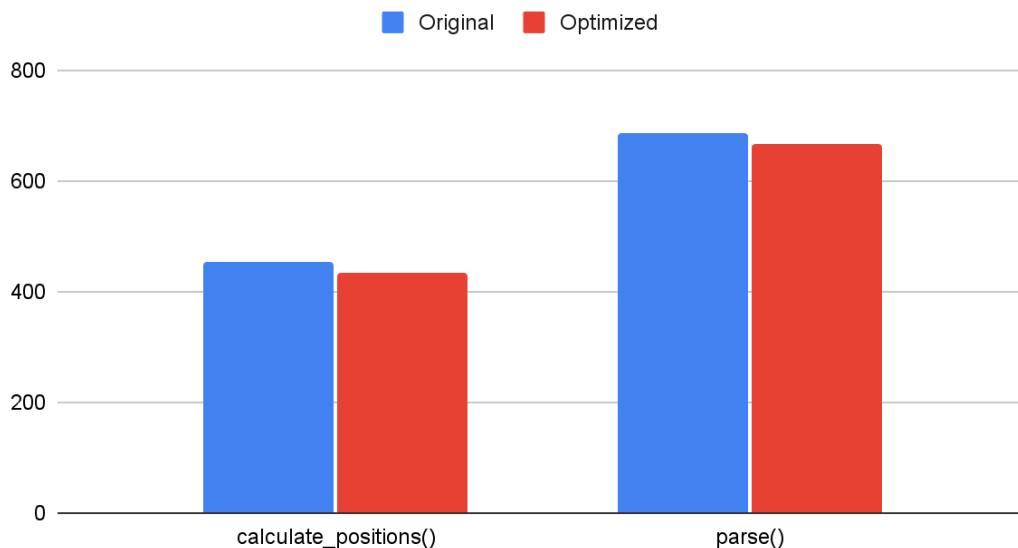
Author: Kathryn Butler

Reviewers: N/A

Source Link: [Commit 53](#)

Supporting Artifact:

Parse IO Adjustment



Comparison of Original and Optimized Functions

Notes: The original code had an object with the data that was written to the file. Then it would re-read the same data to return, which is very inefficient. I adjusted the code so it would return the object instead of reading it back again.

Before optimization:

calculate_positions: 452.7986s
parse: 687.6515s

After optimization:
calculate_positions: 452.7986s
parse: 687.6515s

Improvement: calculate_positions() = 18 seconds faster, parse() = 20 seconds faster.
~3% overall runtime improvement.

The optimized test csv output data was exactly the same compared to the original one.

Vectorized MJS Timestamp Computation

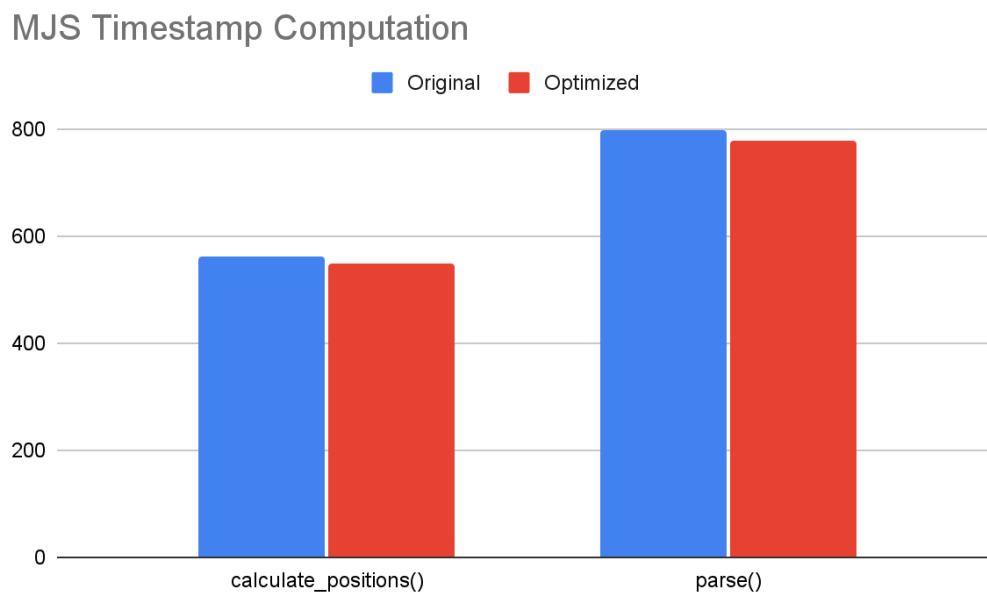
Status: Done

Author: Kathryn Butler

Reviewers: N/A

Source Link: [Issue 14](#)

Supporting Artifact:



Comparison of Original and Optimized Functions

Notes: Refactored MJS logic into a vectorized subfunction, and removed the .apply(axis=1) Python logic for better Pandas functions.

Before optimization:
calculate_positions: 562.7033s

parse: 797.3028s

After optimization:

calculate_positions: 549.5055s

parse: 777.0189s

Improvement: calculate_positions() = 13 seconds faster, parse() = 20 seconds faster.
~2.5% overall runtime improvement.

No runtime crashes after several fixes, and the original output is the same as the optimized one.

Risk and Quality Tracking

Bug Count: 11 ([Link to issue query](#))

Top Bugs:

- [Treating Keys as Positions and Using Argsort](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Series.getitem treating keys as positions is deprecated. Look up how to solve it.)
- [Incorrect conversion from float to datetime](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.)
- [Log.txt hardcoded](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Log.txt seems to be announcing a preset number of cores to calculate. Figure out why it is set to only be 8 cores. Find out if you can manually adjust the core count to be something other than just 8, then add the text to the function dynamically.)
- [Inconsistent timing data](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing

- **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.
- [Xarray.merge conflict](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Set compat explicitly to something other than 'override', then test output for clarity. (Previous goal: Research what "compat" does, and either set it explicitly or use the new combine_kwarg_defaults.)

Next Sprint Plan

- Resolve xarray.merge conflict
 - Investigate and resolve the merge conflict without relying on compat='override'. Compare behavior of join='exact' vs join='outer' using controlled test cases. Source of truth: [Issue 11](#) and [Issue 7](#)
- Split code optimization into individual sections
 - parse() and calculate_positions() responsibility has been divided up. Split these functions into their subproblems and work on them one by one. Update README with new structure and responsibilities. Source of truth: [Issue 13](#)
- Create reproducible testing data from existing dataset
 - Generate a smaller dataset to reproduce runtime inconsistencies. Establish baseline timing benchmarks and document results. Commit dataset to repository. Source of truth: [Issue 4](#)

Team Process Reflection

This sprint felt pretty one-sided in terms of workload. This week there wasn't meaningful progress happening throughout the week, which made collaboration and planning difficult. One teammate has continued to be unresponsive, so at this point we are moving forward without relying on their participation.

We also need to reassess our original goal of achieving a 60% runtime reduction. While we've made measurable improvements and reduced runtime through targeted optimizations, hitting 60% may require deeper changes beyond incremental refactoring. Next sprint, we plan to evaluate whether adjusting the target to something evidence-based makes more sense, while still continuing to optimize as much as possible.